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Salts of novel chelatoborate anions: $[R^FBX(ox)]^-$(R^F = CF_3, C_2F_5; X = F, OMe; ox = oxalato) and $[R^FB(OMe)(cat)]^-$(R^F = CF_3, C_2F_5; cat = catecholato)

Jan A.P. Sprenger [a], Marius Schäfer [a], Nikolai Ignatiev [b], Maik Finze [a,*]

[a] Institut für Anorganische Chemie, Julius-Maximilians-Universität Würzburg, Am Hubland, 97074 Würzburg, Germany
[b] Merck KGaA, PM-ATI, Frankfurter Straße 250, 64293 Darmstadt, Germany

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ABSTRACT

Tetraethylammonium and tetraphenylphosphonium salts of the oxalatoborate anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5B(OCH_3)(ox)]^-$ (**1b**), $[CF_3BF(ox)]^-$ (**1c**), and $[C_2F_5BF(ox)]^-$ (**1d**) have been obtained from easily accessible $K[R^FB(OCH_3)_3]$ and $K[R^FBF_3]$ (R^F = CF_3, C_2F_5), respectively. In addition, the related catecholatoborate anions $[CF_3B(OCH_3)(cat)]^-$ (**2a**) and $[C_2F_5B(OCH_3)(cat)]^-$ (**2b**) have been synthesized starting from $K[R^FB(OCH_3)_3]$ and catechol. Anions **2a** and **2b** have been isolated as $[Ph_4P]^+$ salts. Two different synthetic strategies have been applied: (i) Reaction of the potassium borate with neat oxalic acid or catechol and (ii) reaction of both components in an ethereal solvent. Salts of all six anions have been obtained via these two different routes except for the $[CF_3BF(ox)]^-$ anion (**1c**) that was obtained from a reaction in diglyme, only. The new anions have been characterized by multinuclear NMR spectroscopy and by mass spectrometry ((−)-MALDI or (−)-ESI). Furthermore, the tetraphenylphosphonium salts of the chelatoperfluoroalkylborate anions **1a**, **1d**, and **2a** have been characterized by single-crystal X-ray diffraction.

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1. Introduction

Borate anions are valuable components for materials used in many applications, for example Li^+ salts for battery technologies [1], ionic liquids [2,3], components of electrolytes for dye-sensitized solar cells (Grätzel cells, DSSCs [4]) [5,6], and weakly coordinating anions for the stabilization of highly reactive cations [7–12], e.g. in catalysis. In general, the properties of the anions and therefore of the materials with these anions are determined by the substituents that are bonded to the central boron atom. Many of these simple borate anions have fluorine, alkoxy, or perfluoroalkyl substituents that are bonded to the central boron atom. In addition, a series of borate anions with *O,O′*-donor ligands have been studied, in detail. These include the bis(oxalato)borate anion $[B(ox)_2]^-$ [13–17] and the bis(catecholato)borate anion $[B(cat)_2]^-$ [14,18–21], which are examples for spiroborate anions, as well as the chelatoborate anions $[BF_2(ox)]^-$ [13,22–25] and $[BF_2(cat)]^-$ [26] that contain two fluorine substituents and either one oxalato or catecholato ligand.

Perfluoroalkyl groups often result in improved properties, e.g. chemical and thermal stability, of the borate anions and their salts. Especially, trifluoromethylborate anions and their salts have been studied in detail [27–29]. However, borate anions with CF_3 groups exhibit a lower stability compared to related anions with C_2F_5 groups or longer linear perfluoroalkyl chains [30,31], which limits their applicability. The reason for the reduced stability is the relatively easy abstraction of a fluoride anion to results in a difluorocarbene complex of boron that immediately undergoes further decomposition [30].

In this contribution we report on the synthesis of six new chelatoborate anions that have either an oxalato or a catecholato ligand, a fluoro or a methoxy substituent and one perfluoroalkyl group (CF_3 or C_2F_5) that are bonded to the boron atom. The NMR spectroscopic as well as most of the mass spectrometric data of the anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5B(OCH_3)(ox)]^-$ (**1b**), $[CF_3BF(ox)]^-$ (**1c**), $[C_2F_5BF(ox)]^-$ (**1d**), $[CF_3B(OCH_3)(cat)]^-$ (**2a**), and $[C_2F_5B(OCH_3)(cat)]^-$ (**2b**) are reported [32]. In addition, the crystal

* Corresponding author. Tel.: +49 931 31 85857; fax: +49 931 31 87053.
E-mail address: maik.finze@uni-wuerzburg.de (M. Finze).

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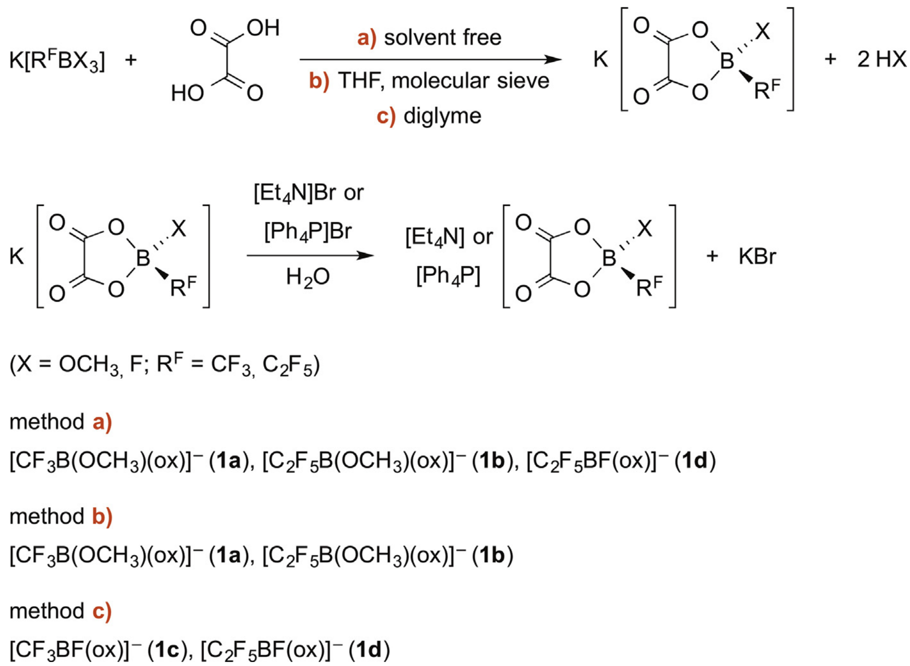


Scheme 1. Syntheses of oxalatoborates.

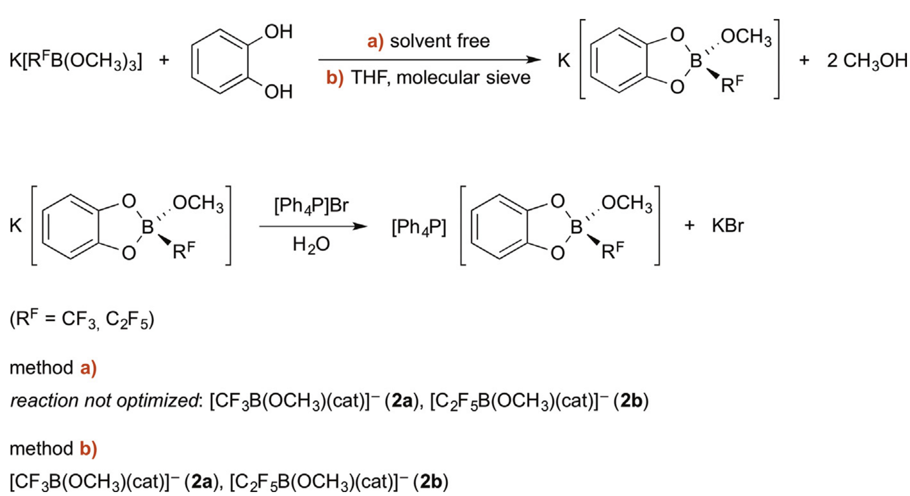
structures of the $[Ph_4P]^+$ salts of the anions **1a**, **1d**, and **2a** are described.

2. Results and discussion

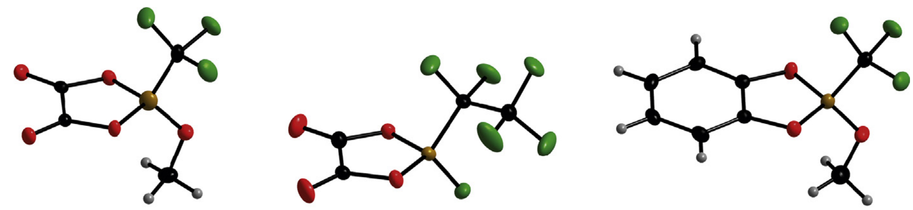
2.1. Syntheses

Tetraethylammonium or tetraphenylphosphonium salts of the oxalatoborate anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5B(OCH_3)(ox)]^-$ (**1b**), and $[C_2F_5BF(ox)]^-$ (**1d**) were obtained by the reaction of the respective potassium perfluoroalkyltrimethoxy or -trifluoroborate with neat oxalic acid followed by metathesis reactions in water with either $[Et_4N]Br$ or $[Ph_4P]Br$ (Scheme 1). The trimethoxyborates react with oxalic acid even at room temperature whereas the corresponding reaction with $K[C_2F_5BF_3]$ was performed at elevated temperatures (175 °C). The methanol that is formed during the reaction is removed under reduced pressure and the released HF evaporates at 175 °C. The reaction of $K[CF_3BF_3]$ with oxalic acid starts at 100 °C. However, no $[CF_3BF(ox)]^-$ (**1c**) was observed and a complex mixture of boron species was obtained. The reason for the decomposition of the $[CF_3BF_3]^-$ anion is the lower stability of trifluoromethyl compared to pentafluoroethyl groups bonded to boron, which is well documented in the literature [30,31].

Salts of the oxalatoborate anions **1a** and **1b** are accessible by stirring of the potassium perfluoroalkyltrimethoxyborates and oxalic acid in THF in the presence of molecular sieve at room temperature, as well (Scheme 1). The molecular sieve captures the methanol that is formed in the course of the reaction. If the methanol is not removed from the reaction mixture large amounts of decomposition products are formed. The respective reaction of $K[R^FBF_3]$ ($R^F = CF_3$, C_2F_5) with oxalic acid does not take place at room temperature or at temperatures below 100 °C. However, heating of $K[R^FBF_3]$ with oxalic acid in diglyme yielded the corresponding anions **1c** and **1d**. The reaction temperature for the preparation of $[CF_3BF(ox)]^-$ (**1c**) in



Scheme 2. Syntheses of catecholatoborates.



$[CF_3B(OCH_3)(ox)]^-$ $[C_2F_5BF(ox)]^-$ $[CF_3B(OCH_3)(cat)]^-$

Fig. 1. The anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5BF(ox)]^-$ (**1d**), and $[CF_3B(OCH_3)(cat)]^-$ (**2a**) in the crystals of their $[Ph_4P]^+$ salts [ellipsoids are drawn at the 30% ($[Ph_4P]$**1a**; $[Ph_4P]$**1d**) and 50% probability level ($[Ph_4P]$**2a**) except for H atoms, which are depicted with arbitrary radii].

diglyme should not exceed 120 °C because at higher temperatures decomposition products are formed. $[C_2F_5BF(ox)]^-$ (**1d**) is more stable than **1c** as outlined above and therefore higher temperatures (boiling diglyme) can be applied for its synthesis.

Catechol reacts with $K[R^FB(OCH_3)_3]$ to yield the corresponding chelatoborate anions $[CF_3B(OCH_3)(cat)]^-$ (**2a**) and $[C_2F_5B(OCH_3)(cat)]^-$ (**2b**), respectively, which were isolated as $[Ph_4P]^+$ salts (Scheme 2). Similar to the syntheses described for the oxalatoborates, the reactions can be performed either with $K[R^FB(OCH_3)_3]$ and neat catechol at room temperature or in THF at room temperature in the presence of molecular sieve. However, only the reactions in THF have been optimized and are therefore described in Section 3. Attempted synthesis of salts of the catecholatofluoroperfluoroalkylborate anions $[CF_3BF(cat)]^-$ and $[C_2F_5BF(cat)]^-$ using the synthetic strategies described herein failed, so far.

The purity of the tetraethylammonium and tetraphenylphosphonium salts of the chelatoborate anions **1a–1d**, **2a**, and **2b** that are obtained via metathesis reactions as outlined in Schemes 1 and 2 are in the range of 80–95% as determined by ^{11}B NMR spectroscopy. Some of the minor products that are formed have been identified by ^{11}B NMR spectroscopy. The spiroborate anions $[B(ox)_2]^-$ [13] (7.5 ppm) and $[B(cat)_2]^-$ [18] (14.2 ppm) are observed in case of reactions with oxalic acid and catechol, respectively. Furthermore, the formation of the $[BF_2(ox)]^-$ anion [13] (3.6 ppm) was unambiguously proven for reactions starting from $K[CF_3BF_3]$ and $K[C_2F_5BF_3]$. For most salts the purity was improved to more than 95% by crystallization from acetone as described in Section 3.

2.2. Crystal structures

The $[Ph_4P]^+$ salts of the chelatoborate anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5BF(ox)]^-$ (**1d**), and $[CF_3B(OCH_3)(cat)]^-$ (**2a**) crystallize in the triclinic space group $P\bar{1}$ with $Z = 2$ (**1a**) and in the monoclinic space groups $C2/c$ (**1d**) and $P2_1/c$ (**2a**) both with $Z = 4$, respectively. The molecular structures of the three related borate anions are depicted in Fig. 1 and in Table 1 selected bond parameters are collected. In the crystals of $[Ph_4P]$**1a** and $[Ph_4P]$**2a** the anions are ordered whereas the $[C_2F_5BF(ox)]^-$ anion (**1d**) is disordered over two positions in the crystal of its $[Ph_4P]^+$ salt. Hence, a detailed discussion of the bond parameters of anion **1d** is arbitrary. However, the related bond parameters of **1d** are close to those found for **1a** and **2a**.

Anions **1a** and **2a** contain a methoxy and a trifluoromethyl group and both of them have an *O,O′*-donor ligand but with different backbones. The $\{BO_3C\}$ moiety of the two related anions reveals a slightly distorted tetrahedral arrangement. The $B–O_{ox}$ distance in **2a** is a little shorter than $d(B–O_{cat})$ in **1a**. In contrast, for $d(B–O_{methoxy})$ and $d(B–CF_3)$ slightly larger values have been determined for **2a** compared to **1a**. These differences indicate small differences in the bonding behavior of the oxalato and the catecholato ligands toward boron. The oxalato ligand reveals a somewhat weaker interaction with boron than the catecholato ligand.

In general, the bond parameters of the three borate anions **1a**, **1d**, and **2a** are similar to the respective bond lengths and angles reported for related simple borate anions, for example in the crystals

Table 1
Selected bond parameters of the anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5BF(ox)]^-$ (**1d**), and $[CF_3B(OCH_3)(cat)]^-$ (**2a**) in the crystals of their $[Ph_4P]^+$ salts.

Anion	**1a**	**1d**	**2a**
Bond lengths (Å)			
$B–O_{ox/cat}$	1.514(3)/1.509(2)	1.487(5)/1.474(5)	1.498(2)/1.498(2)
$B–CF_{2/3}$	1.619(4)	1.637(8)	1.632(2)
B–F	–	1.372(5)	–
$B–OCH_3$	1.399(2)	–	1.431(2)
$O_{ox/cat}–C_{ox/cat}$	1.320(2)/1.318(3)	1.355(7)/1.306(13)	1.358(2)/1.356(2)
$C_{ox/cat}–C_{ox/cat}$	1.535(3)	1.516(10)	1.402(2)
$C_{ox}{=}O$	1.201(3)/1.201(2)	1.212(6)/1.199(13)	–
$BC–F_{2/3}$	1.340(2)–1.359(3)	1.356(7)/1.377(7)	1.367(2)/1.360(2)
$CF_2–CF_3$	–	1.518(14)	–
$C–F_3$	–	1.344(14)–1.320(14)	–
$O–CH_3$	1.417(3)	–	1.410(2)
Bond angles (°)			
$O_{ox/cat}–B–O_{ox/cat}$	102.3(1)	104.6(3)	104.4(1)
$O_{ox/cat}–B–CF_{2/3}$	108.8(2)/107.9(2)	112.3(4)/106.1(4)	110.59(11)/107.75(11)
$O_{ox}–B–F$	–	112.3(3)/111.0(3)	–
$O_{ox/cat}–B–OCH_3$	114.8(2)/114.2(2)	–	114.80(12)/114.21(12)
$CF_{2/3}–B–F$	–	110.4(4)	–
$CF_{2/3}–B–OCH_3$	108.4(2)	–	105.0(1)
$B–O–CH_3$	117.6(2)	–	117.9(1)
$B–CF_2–CF_3$	–	117.2(6)	–

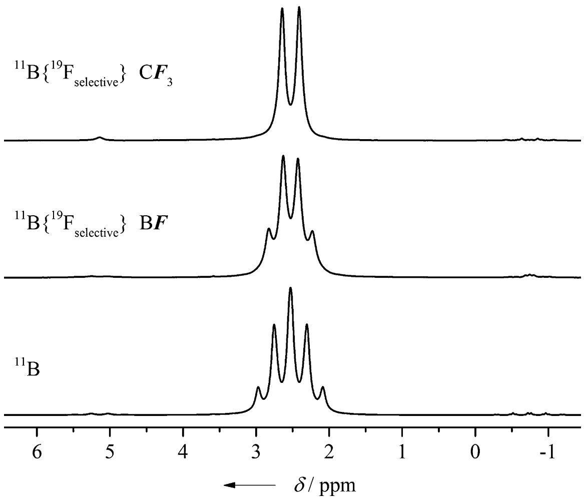


Fig. 2. ^{11}B NMR and $^{11}B\{^{19}F_{selective}\}$ NMR spectra of the $[CF_3BF(ox)]^-$ anion (**1c**).

of $K[CF_3B(OCH_3)_3]\cdot THF$ [33], $K[CF_3BF_3]$ [34], $M[C_2F_5BF_3]$ (M = 1-methyl-2-oxa-5-azoniaspiro[4.4]nonane) [35], $Na[BF_2(ox)]\cdot 2CH_3CN$ [36], $[Me_3PH][B(cat)_2]$ [37], and $Li[B(ox)_2]\cdot H_2O$ [38].

2.3. NMR spectroscopy

The six new simple borate anions **1a**–**1d**, **2a**, and **2b** have been unambiguously characterized by multinuclear NMR spectroscopy and the chemical shifts and coupling constants are summarized in Table 2. The assignment of the signals is straight forward based on the coupling schemes as exemplified by the ^{11}B and $^{11}B\{^{19}F_{selective}\}$ NMR spectra of the $[CF_3BF(ox)]^-$ anion (**1c**) depicted in Fig. 2.

3. Experimental

3.1. General methods

^{1}H, ^{11}B, ^{13}C, and ^{19}F NMR spectra were recorded at 25 °C in $(CD_3)_2CO$ or CD_3CN on a Bruker Avance 500 NMR spectrometer or on a Bruker Avance III HD 300 NMR spectrometer. The NMR signals were referenced against TMS (^{1}H and ^{13}C), $BF_3\cdot OEt_2$ in $CDCl_3$ with $\Xi(^{11}B)$ = 32.083974 MHz and $CFCl_3$ with $\Xi(^{19}F)$ = 94.094011 MHz as external standards. ^{1}H and ^{13}C chemical shifts were calibrated against the residual solvent signal and the solvent signal, respectively ($\delta(^{1}H)$: $(CD_3)(CD_2H)CO$ 2.05 ppm, CD_2HCN 1.94 ppm; $\delta(^{13}C)$: $(CD_3)_2CO$ 206.26 and 29.84 ppm, CD_3CN 118.26 and 1.32 ppm) [39]. MALDI mass spectra were acquired on an Autoflex II LRF (Bruker Daltonics) and ESI mass spectra were recorded on a Waters ZMD Quadrupole mass spectrometer.

3.2. Chemicals

All standard chemicals were obtained from commercial sources and used without any further purification. $K[CF_3BF_3]$ [40], $K[C_2F_5BF_3]$ [41], $K[CF_3B(OCH_3)_3]$ [42], and $K[C_2F_5B(OCH_3)_3]$ [42] were prepared according to literature procedures.

3.3. Single crystal structures

Colorless crystals of the tetraphenylphosphonium salts of the anions $[CF_3B(OCH_3)(ox)]^-$ (**1a**), $[C_2F_5BF(ox)]^-$ (**1d**), and $[CF_3B(OCH_3)(cat)]^-$ (**2a**) suitable for a X-ray diffraction study were grown from acetone by slow evaporation of the solvent. All crystals were investigated with CCD diffractometers using Mo-$K\alpha$ radiation (λ = 0.71073 Å) (Oxford Xcalibur equipped with an EOS detector for $[Ph_4P]$**1a**; Bruker X8-Apex II for $[Ph_4P]$**1d** and $[Ph_4P]$**2a**). All structures were solved by direct methods [43,44] and refinement is based on full-matrix least-squares calculations on F^2 [43,45].

The positions of the hydrogen atoms in the crystal structures were located via ΔF syntheses and refined using idealized bond lengths as well as angles. All non-hydrogen atoms were refined anisotropically. The anion $[C_2F_5BF(ox)]^-$ in the crystal of $[Ph_4P]$**1d** is located on a center of inversion, which results in a disorder of anion **1d** over two positions with equal occupancies. However, the quality of the data was sufficient to enable a refinement of the structure without any restraints.

Calculations were carried out using the ShelXle graphical interface [46]. Molecular structure diagrams were drawn with the program Diamond 3.2i [47]. Experimental details, crystal data, and

Table 2
NMR spectroscopic data of the chelatoborate anions $[R^FB(OCH_3)(ox)]^-$ ($R^F = CF_3$ (**1a**), C_2F_5 (**1b**)), $[R^FBF(ox)]^-$ ($R^F = CF_3$ (**1c**), C_2F_5 (**1d**)), and $[R^FB(OCH_3)(cat)]^-$ ($R^F = CF_3$ (**2a**), C_2F_5 (**2b**))[a].

Anion	1a	1b	1c	1d	2a[b]	2b[b,c]
$\delta(^{11}B)$	2.55	3.41	2.53	2.93	6.22	7.16
$\delta(^{19}F)$ B**F**	–	–	−167.20	−164.20	–	–
$\delta(^{19}F)$ BC**F**$_{2/3}$	−76.00	−135.61	−77.43	−137.50	−74.47	−135.61
$\delta(^{19}F)$ CF_2C**F**$_3$	–	−83.07	–	−83.70	–	−83.50
$\delta(^{13}C)$ C_{ox}	161.32	161.35	160.09	160.09	–	–
$\delta(^{13}C)$ $C_{1,2}$ C_{cat}	–	–	–	–	155.72	154.41
$\delta(^{13}C)$ $C_{3,6}$ C_{cat}	–	–	–	–	108.05	107.92
$\delta(^{13}C)$ $C_{4,5}$ C_{cat}	–	–	–	–	117.12	117.89
$\delta(^{13}C)$ B**C**$F_{2/3}$	131.35	118.52	130.12	117.41	n.o.[d]	n.o.
$\delta(^{13}C)$ CF_2**C**F_3	–	121.98	–	121.77	–	n.o.
$\delta(^{13}C)$ O**C**H_3	47.88	47.90	–	–	47.77	47.73
$\delta(^{1}H)$ OC**H**$_3$	3.13	3.12	–	–	3.06	3.00
$^{1}J(^{19}F, ^{11}B)$	–	–	38.1	41.2	–	–
$^{2}J(^{19}F, ^{11}B)$ $^{11}BC\,^{19}F_{2/3}$	31.7	20.1	34.7	21.2	30.4	17.3
$^{1}J(^{19}F, ^{13}C)$ $B\,^{13}C\,^{19}F_{2/3}$	308.2	258.7	305.6	258.4	n.o.	n.o.
$^{2}J(^{19}F, ^{13}C)$ $B\,^{13}CF_2C\,^{19}F_3$	–	36.8	–	37.4	–	n.o.
$^{1}J(^{19}F, ^{13}C)$ $CF_2\,^{13}C\,^{19}F_3$	–	284.9	–	284.6	–	n.o.
$^{2}J(^{19}F, ^{13}C)$ $C\,^{19}F_2\,^{13}CF_3$	–	32.0	–	31.58	–	n.o.
$^{3}J(^{19}F, ^{13}C)$ $B\,^{19}FCF_2\,^{13}CF_3$	–	–	–	1.4	–	–
$^{2}J(^{19}F, ^{13}C)$ $B\,^{19}F\,^{13}CF_{2/3}$	–	–	58.1	53.7	–	–

[a] δ in ppm, J in Hz; NMR solvent: $(CD_3)_2CO$.
[b] Catechol group: $\delta(^{1}H)$ = 6.37–6.25 ppm.
[c] NMR solvent: CD_3CN.
[d] n.o. = not observed.

Table 3
Selected crystal data and details of the refinement of the crystal structures of $[Ph_4P][CF_3B(OCH_3)(ox)]$ ($[Ph_4P]$**1a**), $[Ph_4P][C_2F_5BF(ox)]$ ($[Ph_4P]$**1d**), and $[Ph_4P][CF_3B(OCH_3)(cat)]$ ($[Ph_4P]$**2a**).

Compound	$[Ph_4P]$**1a**	$[Ph_4P]$**1d**	$[Ph_4P]$**2a**
Empirical formula	$C_{28}H_{23}BF_3O_5P$	$C_{28}H_{20}BF_6O_4P$	$C_{32}H_{27}BF_3O_3P$
Formula weight ($g\,mol^{-1}$)	538.24	576.22	558.32
T (K)	143	100	100
Crystal system	Triclinic	Monoclinic	Monoclinic
Space group	$P\bar{1}$	$C2/c$	$P2_1/c$
a (Å)	11.0744(8)	16.8207(10)	13.0314(9)
b (Å)	11.5314(7)	7.1612(4)	15.3272(10)
c (Å)	12.2196(7)	21.5189(13)	13.4873(9)
α (°)	73.427(5)		
β (°)	69.395(6)	101.981(2)	92.825(2)
γ (°)	61.786(7)		
Volume (Å^3)	1273.7(2)	2535.6(3)	2690.6(3)
Z	2	4	4
D_{calcd} ($Mg\,m^{-3}$)	1.403	1.509	1.378
μ (mm^{-1})	0.168	0.188	0.157
$F(000)$	556	1176	1160
No. of collected rflns	8406	14,857	47,395
No. of unique rflns (R(int))	4482, 0.022	2497, 0.034	5294, 0.043
No. of parameters/restraints	343/0	243/0	362/0
$R1$ ($I > 2\sigma(I)$)	0.038	0.042	0.033
$wR2$ (all)	0.080	0.116	0.087
GOF on F^2	1.092	1.071	1.049
Largest diff. peak/hole/e (Å^{-3})	0.483/−0.324	0.401/−0.234	0.515/−0.308
CCDC no.	1012,998	1012,999	1013,000

CCDC numbers are collected in Table 3. Crystallographic data have been deposited with the Cambridge Crystallographic Data Centre and copies of the data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif.

3.4. Synthetic reactions

*3.4.1. Synthesis of $[Et_4N][CF_3B(OCH_3)(ox)]$ ($[Et_4N]$**1a**)*

$K[CF_3B(OMe)_3]$ (1.0 g, 4.72 mmol) and anhydrous oxalic acid (0.8 g, 8.89 mmol) were placed into a reaction vessel. After 15 min at room temperature of stirring the solid liquefied. The methanol that had formed was removed in a vacuum. The residue was dissolved in water (10 mL) and $[Et_4N]Br$ (1.5 g, 7.14 mmol) was added. The reaction mixture was extracted with CH_2Cl_2 (2×10 mL). The combined organic layers were dried with $MgSO_4$ and the solvent was removed under reduced pressure. Colorless $[Et_4N]$**1a** was isolated that contained approximately 10% of other tetraethylammonium borates. Yield: 1.03 g, 3.04 mmol, 64%. Pure $[Et_4N]$**1a** was obtained by recrystallization from acetone.

*3.4.2. Synthesis of $[Et_4N][C_2F_5B(OCH_3)(ox)]$ ($[Et_4N]$**1b**)*

$[Et_4N]$**1b** was synthesized according to the method described for $[Et_4N]$**1a** using $K[C_2F_5B(OMe)_3]$ (1.0 g, 3.82 mmol), anhydrous oxalic acid (0.7 g, 7.78 mmol), and $[Et_4N]Br$ (1.5 g, 7.14 mmol). The purity of the crude product was >90% according to the ^{11}B NMR spectra. Yield: 0.69 g, 1.82 mmol, 47%. Recrystallization of $[Et_4N]$**1b** afforded pure product.

*3.4.3. Synthesis of $[Ph_4P][CF_3B(OCH_3)(ox)]$ ($[Ph_4P]$**1a**)*

To anhydrous oxalic acid (110 mg, 1.2 mmol) and molecular sieve (4 Å) a solution of $K[CF_3B(OMe)_3]$ (250 mg, 1.2 mmol) in THF (20 mL) was added and the reaction mixture was stirred for 20 h at room temperature. The molecular sieve was filtered off and the THF was removed in vacuo. The solid residue was treated with a solution of $[Ph_4P]Br$ (550 mg, 1.3 mmol) in water (40 mL). Immediately a solid precipitate formed that was filtered off and dried under reduced pressure. Yield: 370 mg, 0.69 mol, 58%. (−)-ESI-MS m/z calcd for **1a** ($[C_4H_3BF_3O_5]^-$): 199(100%), 198(25%), 200(5%), 201(1%). Found: 199(100%), 198(21%), 200(6%), 201(2%).

*3.4.4. Synthesis of $[Ph_4P][C_2F_5B(OCH_3)(ox)]$ ($[Ph_4P]$**1b**)*

$[Ph_4P]$**1b** was synthesized according to the method described for $[Ph_4P]$**1a** starting from $K[C_2F_5B(OMe)_3]$ (320 mg, 1.2 mmol), anhydrous oxalic acid (110 mg, 1.2 mmol), and $[Ph_4P]Br$ (510 mg, 1.2 mmol). Yield: 320 mg, 0.54 mmol, 45%. (−)-MALDI-MS m/z calcd for **1b** ($[C_5H_3BF_5O_5]^-$): 249(100%), 248(25%), 250(6%). Found: 249(100%), 248(19%), 250(5%).

*3.4.5. Synthesis of $[Et_4N][CF_3BF(ox)]$ ($[Et_4N]$**1c**)*

$K[CF_3BF_3]$ (500 mg, 2.84 mmol) and anhydrous oxalic acid (2.5 g, 27.77 mmol) were dissolved in diglyme (40 mL) and the reaction mixture was stirred at 120 °C for 6 h. All volatiles were removed in a vacuum and the residue was suspended in water (50 mL). The insoluble material was filtered off and the solution was treated with solid $[Et_4N]Br$ (1.0 g, 4.76 mmol) and then extracted with CH_2Cl_2 (3×10 mL). The combined organic layers were dried with $MgSO_4$ and the solvent was removed under reduced pressure. Colorless $[Et_4N]$**1c** was isolated that contained approximately 15% of other tetraethylammonium borates. Yield: 310 mg, 0.98 mmol, 34%. One of the byproducts of the reaction was found to be $[Et_4N][B(ox)_2]$ by ^{11}B NMR spectroscopy. A differentiation between the anions **1c** and $[B(ox)_2]^-$, which have almost the same mass, was not possible by mass spectrometry.

*3.4.6. Synthesis of $[Ph_4P][C_2F_5BF(ox)]$ ($[Ph_4P]$**1d**)*

A mixture of anhydrous oxalic acid (260 mg, 2.9 mmol) and $K[C_2F_5BF_3]$ (200 mg, 0.88 mmol) in diglyme (20 mL) was heated to reflux for 19 h. All volatiles were removed in a vacuum. The solid residue was taken up into in water (50 mL) and an aqueous solution of $[Ph_4P]Br$ (0.9 g, 2.1 mmol, 100 mL) was added. The colorless precipitate (400 mg) that was filtered off was a mixture of 75% of $[Ph_4P][C_2F_5BF_3]$ and only 15% of $[Ph_4P]$**1d**. Recrystallization from acetone gave a few crystals of $[Ph_4P]$**1d**. (−)-MALDI-MS m/z calcd for **1a** ($[C_4BF_6O_4]^-$): 237(100%), 236(25%), 238(5%). Found: 237(100%), 236(26%), 238(6%).

*3.4.7. Synthesis of $[Et_4N][C_2F_5BF(ox)]$ ($[Et_4N]$**1d**)*

A mixture of anhydrous oxalic acid (330 mg, 3.67 mmol) and $K[C_2F_5BF_3]$ (220 mg, 0.97 mmol) was heated to 175 °C for about 5 min and the gaseous HF that formed instantaneously was removed under reduced pressure. The solid residue was dissolved in water (10 mL) and solid $[Et_4N]Br$ (300 mg, 1.43 mmol) was added. The reaction mixture was extracted with CH_2Cl_2 (3×10 mL). The combined organic layers were dried with $MgSO_4$ and the solvent was removed under reduced pressure. The colorless crude $[Et_4N]$**1d** contained approximately 10% of other tetraethylammonium salts of other borate anions. Yield: 280 mg, 0.76 mmol, 78%.

3.4.8. Synthesis of $[Ph_4P][CF_3B(OCH_3)(cat)]$ ($[Ph_4P]$**2a**)

$[Ph_4P]$**2a** was synthesized analogously to $[Ph_4P]$**1a**. $K[CF_3B(OMe)_3]$ (250 mg, 1.2 mmol), catechol (110 mg, 1.0 mmol), and $[Ph_4P]Br$ (550 mg, 1.3 mmol). According to the ^{11}B NMR spectra, the crude colorless product contained approximately 15% of other $[Ph_4P]^+$ borates. Yield: 250 mg, 0.45 mmol, 45%. Pure $[Ph_4P]$**2a** was obtained by crystallization from acetone. (−)-ESI-MS m/z calcd for **2a** ($[C_8H_7BF_3O_3]^-$): 219(100%), 218(25%), 220(9%). Found: 219(100%), 218(22%), 220(10%).

3.4.9. Synthesis of $[Ph_4P][C_2F_5B(OCH_3)(cat)]$ ($[Ph_4P]$**2b**)

$[Ph_4P]$**2b** was synthesized as described for $[Ph_4P]$**1a** starting from $K[C_2F_5B(OMe)_3]$ (500 mg, 1.9 mmol), catechol (210 mg, 1.9 mmol), and $[Ph_4P]Br$ (2.8 g, 1.3 mmol). The solid material contained approximately 15% of other $[Ph_4P]^+$ borates. Yield: 310 mg, 0.50 mmol, 26%. Crystallization from acetone afforded pure $[Ph_4P]$**2b**. (−)-ESI-MS m/z calcd for **2b** ($[C_9H_7BF_5O_3]^-$): 269(100%), 268(24%), 270(10%). Found: 269(100%), 268(28%), 270(15%).

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